AL-2002 - Artificial Intelligence Lab
Lab # 7

Objectives:

- Propositional Logic

Note: Carefully read the following instructions (*Each instruction contains a weightage*)

1. First think about statement problems and then write your program.
2. Write Program in multiple cells of the same notebook**.**
3. **Must Use Functions and Exception handling for each question, where they can be applied.**
4. Do not copy from any source otherwise you will be penalized with negative marks.
5. Add your source code in word document and attach output screenshots.
6. Upload word file and ipynb file.
7. Please submit your **Both files** with this naming convention ROLLNO_SECTION_LABNO. **(Do not upload zip file).**
8. Submit your lab on Google Classroom.

Propositional logic, also known as sentential logic or propositional calculus, is a branch of formal logic that deals with propositions, which are statements that can either be true or false but not both. Propositional logic studies how these propositions can be combined using logical operators to form more complex statements (sentences).

The **syntax** of propositional logic defines the allowable sentences. Complex sentences are constructed from simpler sentences, using parentheses and operators called **logical connectives**.

There are five connectives in common use, which hold for any subsentences P and Q (atomic or complex) in any model m (here "iff" means "if and only if"):

- ¬, **not**: ¬P is true if and only if (iff) P is false in m. The atomic sentences P and ¬P are referred to as **literals**.
- ∧, **and**: P ∧ Q is true iff both P is true and Q is true in m. An 'and' sentence is known as a **conjunction** and its component propositions the **conjuncts**.
- ∨, **or**: P ∨ Q is true iff either P is true or Q is true in m. An 'or' sentence is known as a **disjunction** and its component propositions the **disjuncts**.
- ⇒, **implication**: P ⇒ Q is true unless P is true and Q is false in m.
- ⇔, **biconditional**: P ⇔ Q is true iff either both P and Q are true or both are false in m.

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$P \Leftrightarrow Q$
false	*false*	*true*	*false*	*false*	*true*	*true*
false	*true*	*true*	*false*	*true*	*true*	*false*
true	*false*	*false*	*false*	*true*	*false*	*false*
true	*true*	*false*	*true*	*true*	*true*	*true*

Question 1:
Implement the truth table above.

Note: You can get aid from https://ggc-discrete-math.github.io/logic.html

Also, Implement the following truth table by employing **the capabilities of logic.py library** provided by AIMA repository

Question 2:
Inference rules are the templates for generating valid arguments. Inference rules are applied to derive proofs in artificial intelligence, and the proof is a sequence of the conclusion that leads to the desired goal.

Apply the inference rule to get the below Conclusions from the Truth Table given in Task 1.

Example 1:
Statement-1: "If I am sleepy then I go to bed" ==> P → Q
Statement-2: "I do not go to the bed."==> ~Q
Conclusion: Which infers that "I am not sleepy" ==> ?

Example 2:
Statement-1: Today is Sunday or Monday. ==>P ∨ Q
Statement-2: Today is not Sunday. ==> ¬P
Conclusion: Which infers that "Today is Monday" ==> ?

Example 3:
Statement-1: I have a vanilla ice-cream. ==> P
Statement-2: I have Chocolate ice-cream. ==> Q.
Conclusion: Which infers that "I have vanilla or chocolate ice-cream" ==> ?